Mushfique Shahriar Aurnab

UNDERGRADUATE RESEARCHER / DGIST

mushfique@dgist.ac.kr / github.com/mushfique-dgist / Daegu, Republic of Korea

RESEARCH INTERESTS Wearable and embodied intelligence / multimodal sensing / representation learning / continual adaptation / tool-using AI systems

EDUCATION

Daegu Gyeongbuk Institute of Science and Technology (DGIST) — Aug 2024 - Present

B.S. student, School of Undergraduate Studies

Selected coursework: Introduction to Probability and Mathematical Statistics with R, Artificial Intelligence Basics, Object-Oriented Programming, Data Structure, System Programming

RESEARCH EXPERIENCE

Undergraduate Researcher, UGRP I — Spring 2026

A Comparative Study of Catastrophic Forgetting in Vision Transformers and Convolutional Neural Networks

- Designed and ran a nine-run sequential CIFAR-10 diagnostic (3 class splits x 3 seeds) to separate classifier-readout loss from deeper representation loss using linear probes at six depths.
- At Task-B epoch 25, 79.7% of misclassified old-task examples remained linearly recoverable; refitting a full linear readout restored mean old-task accuracy to 74.6%.
- Evaluated five replay-selection strategies and reported the negative result: random replay outperformed collision-, fate-, memorization-, and speed-based targeting on buffer-only recovery.

SELECTED PROJECTS

GPX Motion Analysis / *Python / time-series alignment / Matplotlib* — **2026**

- Rebuilt a 2025 course project as a tested CLI for gap-aware alignment of asynchronous GPX tracks; exports motion metrics, missingness provenance, and stop-threshold sensitivity from synthetic fixtures.

Vox Pop / *Python / MCP / FastEmbed* — **2026**

- Built a Python package, CLI, and MCP server that searches nine public discussion platforms, routing queries through agent hints, optional LLM rewriting, local embeddings, and a broad fallback.

Course Extractor / *Python / Requests / BeautifulSoup* — **2026**

- Built a catalog scraper with automatic FOSE API discovery plus JSON-LD and HTML fallbacks; validated structured JSON and Markdown output on 508 Harvard DCE courses with zero reported page errors.

MapleStory Code Battle AI / *Python / OpenCV / minimax* — **2025**

- Converted screen captures into board state using template matching and HSV ownership detection, then selected moves with summed-area tables and 3-ply minimax with alpha-beta pruning.

TECHNICAL SKILLS

Languages	Python, C++
ML and vision	PyTorch, torchvision, NumPy, OpenCV, linear probing
Systems and data	Git, asynchronous HTTP APIs, MCP, CLI packaging, JSON/Markdown pipelines